

Formulate First

Inverse problems should be posed first in function space.
 Discretization should approximate the problem, not redefine it.

The Elements

- Starting with a basis is convenient, but it can hide modelling assumptions.
- Instead we should start by defining the model space $\mathcal{M}$, data space $\mathcal{D}$, forward map G , and a description of uncertainty and prior information (1,2) - not by discretization.

Example Problem Setup

We infer a model from indirect, noisy observations.

$$\tilde{\mathbf{d}} = G(m) + \epsilon \quad [G(m)]_i = \int_0^{6371} K_i(x)m(x)dx$$

- We assume the model space to be a function space.
- We place a Gaussian prior on the model space $\mathcal{N}(m_0, C_0)$
- We assume Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, C_D)$
- We solve for the posterior using Bayesian inversion, which has the closed form solution (3):

$$C_M = C_0 - C_0 G^* (G C_0 G^* + C_D)^{-1} G C_0$$

$$\tilde{m} = m_0 + C_0 G^* (G C_0 G^* + C_D)^{-1} (\tilde{\mathbf{d}} - G(m_0))$$

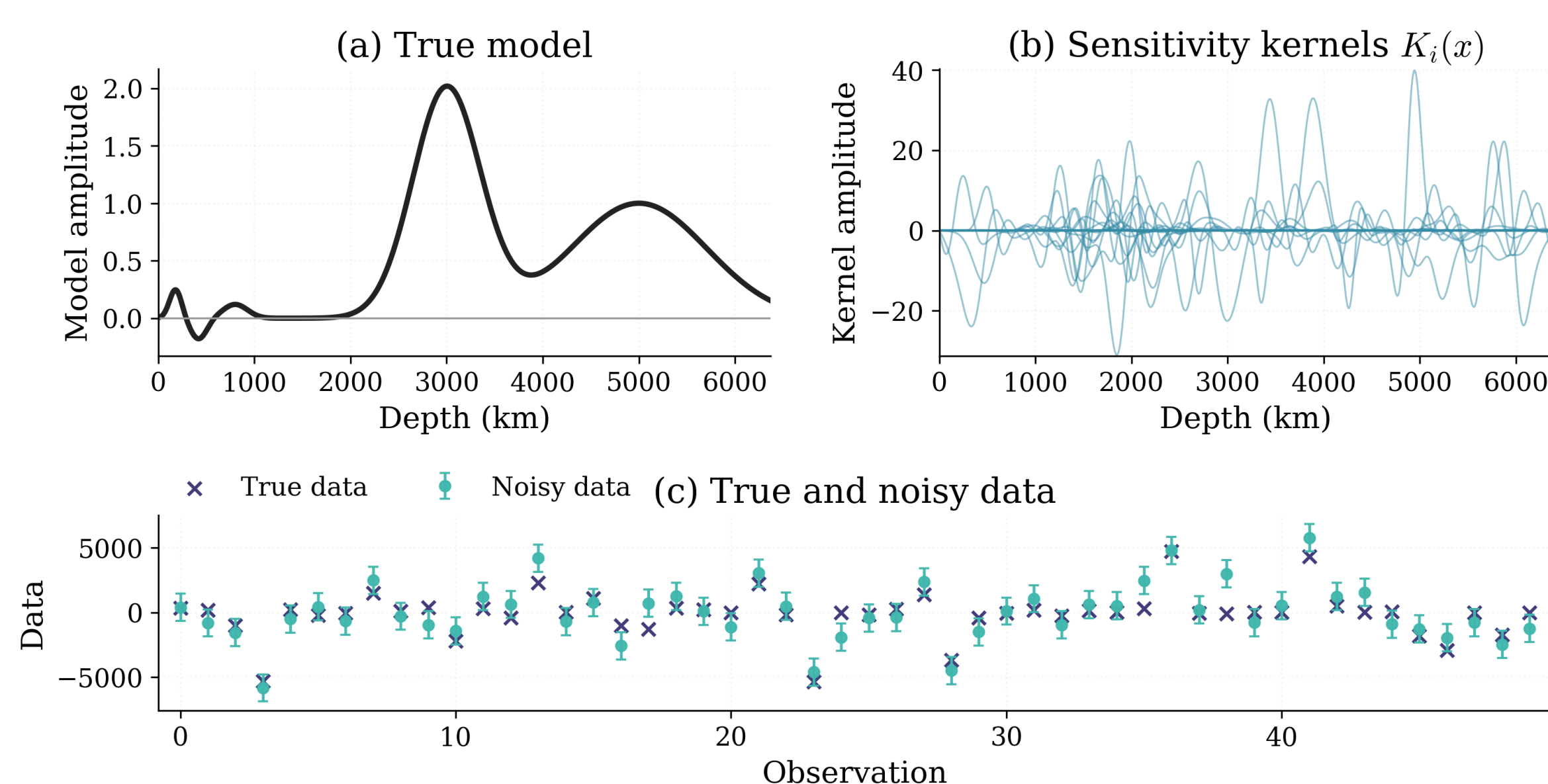


Fig. 1: Toy inverse problem setup: the true model, the sensitivity kernels of the forward operator, and the corresponding noisy observations

Discretization Diagram

Coefficient space	$\mathbb{V}_N$	Finite-dimensional discretized model space
Analysis operator	Π	Maps model to coefficient vector
Synthesis map	π	Reconstructs model from coefficients
Continuous forward	G	Infinite-dimensional forward map
Discrete forward	$\mathbf{G}$	Matrix approximation of G
Prior cov. (cont.)	C_0	Continuous prior covariance operator
Prior cov. (disc.)	$\mathbf{C}_0$	Discretized covariance matrix
Posterior cov (cont.)	C_M	Cont. posterior covariance operator
Posterior cov (disc.)	$\mathbf{C}_{V_N}$	Discretized posterior covariance matrix
Data cov.	$\mathbf{C}_D$	Data covariance matrix
Mass matrix	$\mathbf{M}$	Inner product matrix for $\mathbb{V}_N$

Table: Summary of symbols

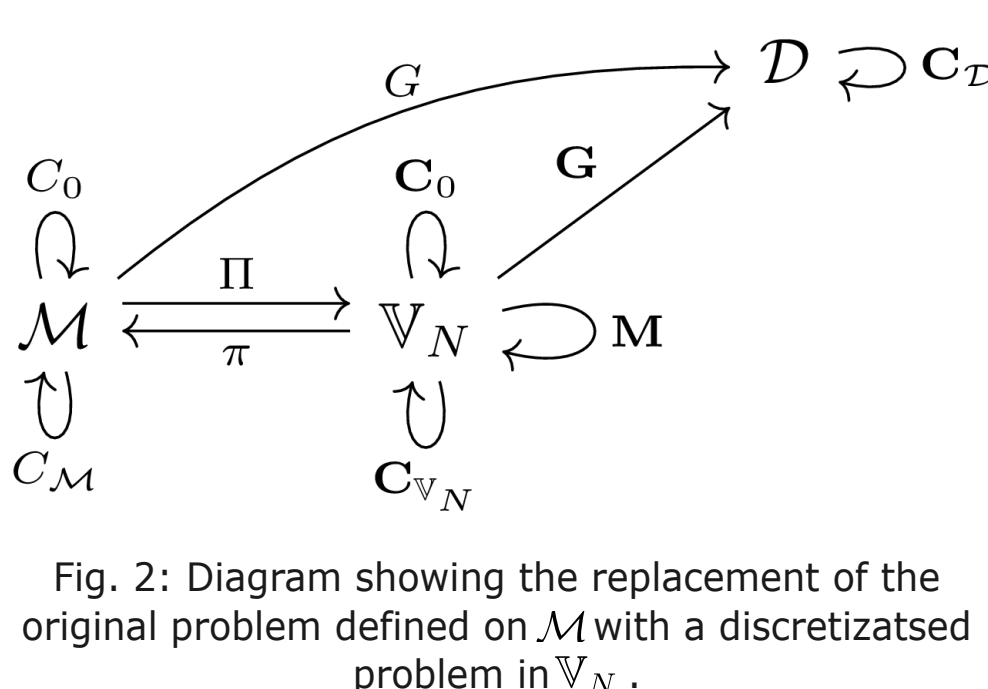


Fig. 2: Diagram showing the replacement of the original problem defined on $\mathcal{M}$ with a discretized problem in $\mathbb{V}_N$.

Your Prior Is Not Just a Matrix

Naive Diagonals

Not every matrix defines a valid covariance operator in the continuous limit (3). For example, a scaled identity is not a good choice for a prior covariance! (3,4) As numerical resolution increases:

- Naive diagonals ($\sigma^2 \mathbf{I}$) $\rightarrow$ diverging variance
- Proper covariances $\rightarrow$ finite variance

This leads to unstable vs stable posteriors.

Therefore, it is crucial a covariance must come from a valid operator.

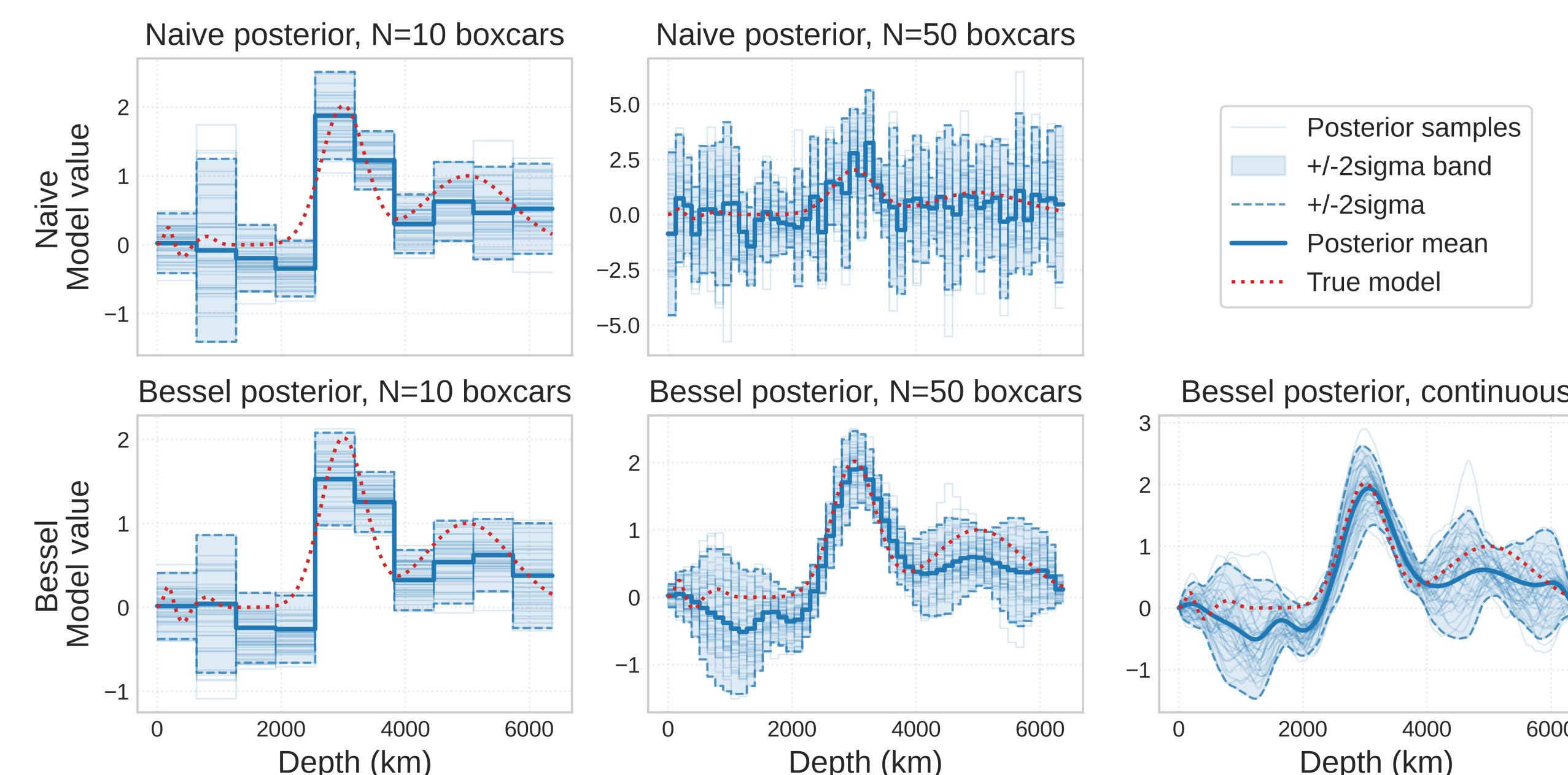


Fig. 3: Naive discretization becomes increasingly unstable with refinement (variance grows from $N=10$ to $N=50$), whereas the correctly discretized Bessel posterior converges toward the continuous solution with controlled uncertainty.

Boundary Conditions

- Covariances in function spaces are operators and may require boundary conditions (3), which are part of the prior information.
- Discretization must respect this structure.

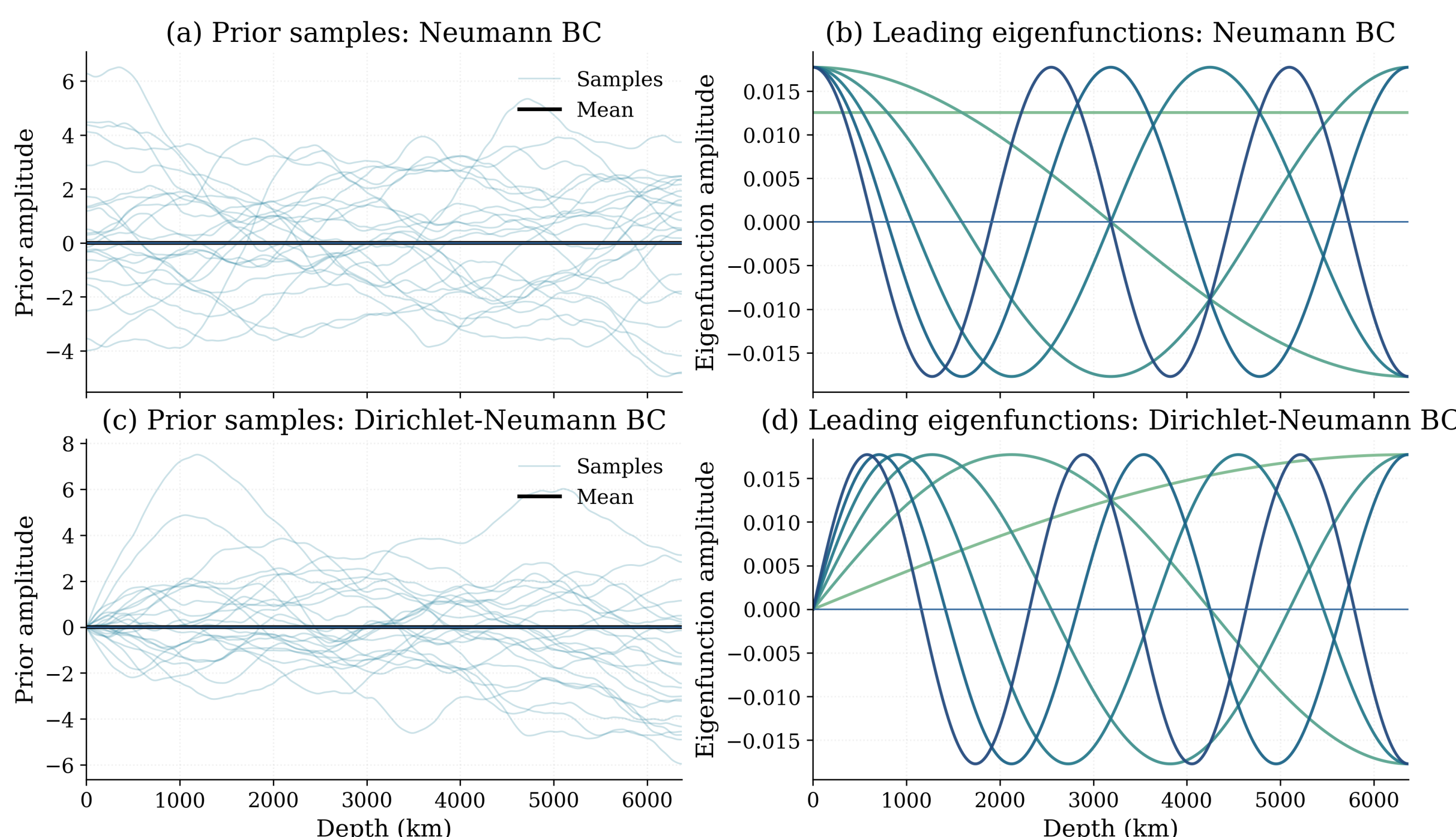


Fig. 4: Continuous Bessel priors under two boundary conditions: posterior-consistent sample realizations are shown for Neumann (top) and Dirichlet-Neumann (bottom), with the black curve indicating the zero mean profile across depth.

Consistent Geometry

Non-orthonormal bases induce a geometry on the discretized model space. The inner product on the discretized model must be weighted by the gram matrix of the basis. Adjoint must be defined with respect to this geometry (5). Ignoring it changes the problem and the solution.

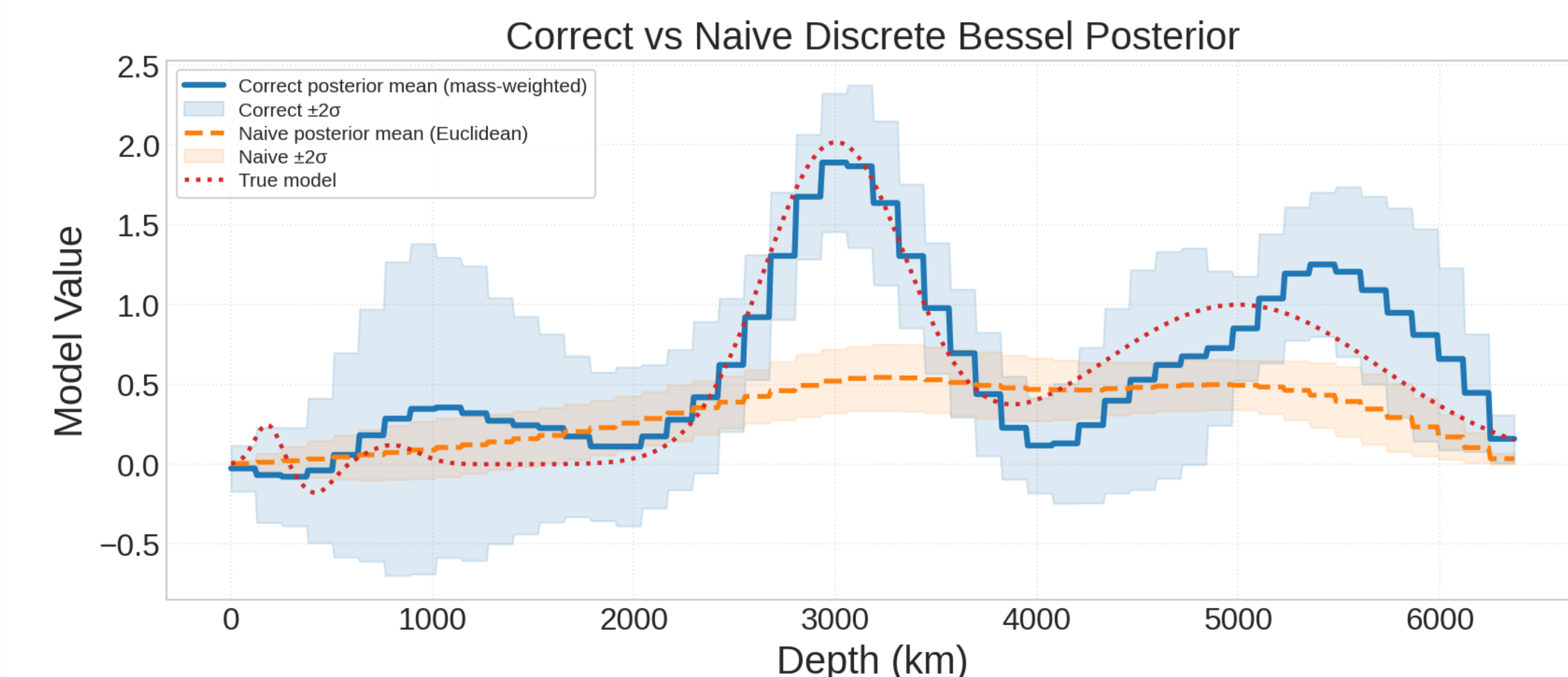


Fig. 5: Geometry-aware vs naive discretized Bayesian posterior.

Conclusions

- Discretization can hide modelling choices.
- Formulate the problem before discretizing it
- Sound workflows make these choices explicit
- These ideas are implemented in the pygeoinf package (scan QR).



pygeoinf: python package for geophysical inversions and inferences built on function analytical principles with discretization agnostic implementation



Personal website



intervalinf: modular extension of pygeoinf for function spaces defined over a 1D interval

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